

SUDESH MITHAPELLI

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ABOUT ME

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📍 Pune - 411044

ACADEMIC BACKGROUND

Master's in Modeling and Simulation

(Computational Fluid Dynamics)

Center for Modeling and Simulation

Pune University GPA - 4.08/6

Project - Numerical Modeling of a Fixed Bed Adsorber

Bachelors in Mechanical Engineering

Pune University 1st class

Project - Design and Manufacturing of work holding fixtures

PROFICIENT SKILLS

- CFD Tools - ANSYS, FLOW3D, FLOW3D CAST, FLOW3D WELD, FlowSight, Enight
- Meshing Tools - ANSA, GAMBIT
- CAD Tools - RHINO, CATIA, I-DEAS, SOLIDWORKS, AUTOCAD, AutoLISP.
- Programming - PYTHON, MATLAB, C, C++ 11/14/17/20, Data Structures and Algorithms, STL
- Operating Systems - WINDOWS, LINUX
- Customer Support platform - Salesforce
- Azure DevOps, CMake

LANGUAGES FAMILIAR

English, Hindi, Marathi, Telugu

KEY AREAS WORKED ON

C++ Development, CFD, CAE, Heat transfer, Fluid Mechanics, Fluid Dynamics, Water, Hydraulics, Casting, Turbulence modeling, PyQt based Flow Sight (Post-processor and visualisation tool) GUI development, Technical support, Embedded, Linux driver programming, TCP/IP, Socket programming, Multi-threading, Git, OOPS, OOAD, gRPC, CMake.

INTERESTS

Long Distance Running, Ultra-marathon, Marathon

SUMMARY

A dynamic, diligent, performance driven and highly motivated Engineer with 10 years of experience which includes Python Qt based post-processor GUI development, Python Automation for FlowSight and FLOW3D features in CFD/CAE domain, C++ Development and Automation for Telecommunication applications. Experience in Embedded, Linux driver programming, UART based serial communication, TCP/IP based Socket programming, Multi-threading

WORK EXPERIENCE

C/C++ Software Development - Jan 2024 to present
Emerson, Pune

- Embedded Linux Drivers programming(UART).
- TCP/IP Socket programming.
- C/C++ development, Python scripting.
- Developing Simulators for DeltaV product related applications.

Technical Lead (Mar 2020 to Jan 2024)

Tech Mahindra Pvt. Ltd., Pune

- Telecommunication application
- Lead in various modules
- C++ development, Python scripting & automation

CFD Engineer - Nov 2014 to Nov 2019

Flow Science Software Pvt. Ltd. – Pune

- Python Qt based FlowSight (Post-processor and Visualisation tool) GUI development.
- FLOW3D Customization in C++ and FORTRAN
- Develop and deliver training for FLOW3D CFD/CAE products to the customers.
- Develop test problems and procedures to ensure continuous improvements to FLOW3D CFD/CAE products.
- CFD/CAE products Solver Testing.

CFD Engineer - Aug 2013 to Nov 2014

Aprameya Associates – Pune

- CAD Modeling
- CFD Analysis of Electrostatic Precipitator & Bag house Filter.
- Numerical Simulation of Torpedo Firing Mechanism.

PROJECTS WORKED ON

★ EMERSON

- Simulator(serial communication Linux based)
 - Designing and developing from scratch a serial communication UART protocol based Simulator that Simulates/mimics IO cards.
 - This is developed in C/C++ on Linux platform using Linux driver programming.
 - It has two parts: Application and the Kernel side code that maintains respective databases and update each other by opening and closing character devices.
- Simulator(Network communication Windows based)
 - Designing and developing from scratch a Network communication(TCP/IP) based Simulator that Simulates/mimics IO cards.
 - This is developed in C/C++ on Windows and involves reading data from a Gateway application SQLite database, receiving messages from Client and processing and sending responses from the Server end via TCP/IP communication(Socket programming).
 - Implementing gRPC protocol for Interfacing with Simulation web application.

★ TECH MAHINDRA PRIVATE LIMITED

- Write programs and applications for a business unit on a C/C++ program.
- Guide junior colleagues for developing code as per the requirement specified.
- Perform testing, debugging and fixing applications using the C/C++ platform.
- Design, develop, test, implement and code solutions on C/C++ programs on windows and Unix.
- Review, analyse and assess programs, designs and specifications for own programs on C/C++ applications.
- Create integrated features on payroll enterprise/Telecommunication domain software solutions on SQL, Oracle, and other C/C++ programs.
- Communicate designs and implementations to the team members and the documentation department.
- Complete appropriate test plans and documentation using the C/C++ program.
- Perform modifications and break fixes with C/C++ program.
- Quickly analyse and interpret code through performing system support analysis.
- Complete projects on time and within the budget limit Ability to work and resolve issues independently.
- SVN to Git migration activity.
- Shell scripting.
- Support Testing team and Application Support Team.
- TCP/IP based Socket programming for various applications associated with the Telecommunication domain.

★ FLOW SCIENCE SOFTWARE PRIVATE LIMITED

- **Python Qt based FlowSight(Post-processor and Visualization tool) GUI development in Windows & Linux.**

- Develop various features and make them available in FlowSight GUI, like Right click menus.
 - Add Modify dialogs for existing features.
 - Add new widgets and inculcate new features through it.
 - Modify existing features.
 - Coordinate with the Testing team to train them for Testing the post-processing features available in FlowSight.
 - Automate and add processes like FFT calculations at the flick of a button in FlowSight GUI using Python numpy library.
- **Python Automation(Windows)**
- Automate generation of material data files using Python for FLOW3D Material database.
 - Set up or write automated test scripts in Python and Squish for GUI Testing.
 - Modify existing Python test scripts as per the changes in the GUI.
- **CFD/CAE Solver Testing(Windows & Linux)**
- Testing of the new models developed and about to be inculcated in the next cycle of FLOW-3D release.
 - Collaborating with GUI and Solver developer teams.
- **Continually learn a wide range of physics and numerical methods applied to existing and newly developed models.**
- Learn, train and teach new Physics and Numerics based models.
- **Develop test problems and procedures to ensure continuous improvements to FLOW-3D CFD /CAE products.**
- Know problems faced by customers in the current FLOW3D version.
 - Optimise and set up small test cases to help Developers understand the problem.
 - Meanwhile, suggest alternatives to the customers to get their job done.
 - Provide improvised solutions to the customers after thorough Testing.
- **Collaborate with the Marketing and Sales team to demonstrate and promote the products to the potential customers.**
- Set up test cases as a part of Proof of concept for demonstration purposes.
 - Set up automated running of Simulations in batch mode for conferences.
 - Develop test cases continuously on Verification and Validation projects to create a database for showcase purposes.
- **Develop and deliver training for FLOW-3D to the customers either in-house or on- site for a better and efficient usage of FLOW-3D CFD/CAE products.**
- Creating FlowSight recordings for uploading on the User site.
 - Updating FLOW3D, FlowSight user manual as per the release work.
- **FLOW3D Customization (in C++)**
- Updating FLOW3D, FlowSight user manual as per the release work.
 - Implement customizations for FLOW3D customers in C++ and FORTRAN and try inculcating the new customizations as a new functionality in the future release.

➤ **Provide Technical support and guidance to the customers with their complex and challenging 3D simulations in FLOW-3D CFD/CAE products.**

- Client onboarding
- Troubleshooting
- Efficient simulation setup
- Correct usage of the Physics models available in FLOW3D.
- Efficient post-processing
- Training via live video meetings, emails or on Telephone.

★ **APRAMEYA ASSOCIATES**

➤ **CFD Analysis of Electrostatic Precipitators**

- Receiving Inputs from client
- Reading 2D drawings in AutoCAD
- Creating 3D models in I-DEAS.
- Meshing in GAMBIT.
- Pre-process, Run Simulations and Post-process in ANSYS CFX.
- Implement design variations for various flow guiding vanes.
- Prepare presentations to communicate results from Analysis to the customers.

➤ **Numerical Simulation of Torpedo firing mechanism on Windows.**

- Develop an optimised numerical tool which will suggest Initial Pressure for firing Torpedo from submarine as per the depth of submarine operation.
- Programming in MATLAB.

➤ **Developing User defined functions in C++ for Simulations.**

❖ **M.TECH PROJECTS**

➤ **Develop a Numerical Model for a Fixed Bed Adsorber(Final year project) on Windows.**

- Developed Numerical model that mimics behaviour of a fixed bed adsorber used in water filter cartridges.
- Programming and Computing in MATLAB.
- Discretized Partial difference equations and the CFD techniques to solve the governing equation of the model.
- Solved equations using both Implicit and Explicit methods.

➤ **Implemented Game theory Application on Windows and Linux based environments.**

- Programming in C++.
- Object oriented programming.

➤ **Implement Numerical methods in C++ (Windows & Linux environment)**

❖ **B.E. PROJECT**

➤ **Design and Manufacturing of work holding fixtures.**

- Developed a new light weight fixture design.
- Used ALUMEC for reduction in weight.
- Provided optimised clamping.
- Reduced work load on the pneumatic systems.

Certifications done:

Data Structures and Algorithms

Source - Udemy

Programming in C++

Source - NPTEL(IIT MADRAS)

Modern C++ 11/14/17

Source - Udemy